

SHADOW ESTIMATOR — ELECTRICAL EDITION

Electrical Project Status Summary, Technical Backlog & Deliverables Timeline

Project Name: Shadow Estimator (Dedicated Electrical Blueprint Takeoff)	Current Date: August 2026
Architecture: Next.js 14 App Router, React 18, `@google/genai` (Gemini 2.13)	Primary Trade Focus: Electrical (Power, Lighting, Gear & Conduit)

1. Executive Overview & Electrical Trade Focus

Shadow Estimator is an AI-assisted blueprint takeoff and material estimating platform repurposed to focus **solely on Electrical construction**. It converts raw electrical plan drawings (power plans, lighting layouts, panel schedules, single-line diagrams) into structured line-item electrical estimates by pairing computer vision AI with electrical material catalogs, conduit tools, and automated accessory rules.

4,200+ Electrical Materials Conduit, Wire, Outlets, Panels	Gemini 2.13 Electrical AI Engine Receptacle & Lighting takeoff	Conduit & MC Linear Feet Measurement EMT, PVC, Romex & MC Cable	Instant Box & Strap Rules Auto J-Box & fitting calculation
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2. Operational Core Modules & Electrical Adaptation

A. Electrical Blueprint Takeoff Engine ([TakeoffViewer.js](#), [/api/takeoff/convert](#))

- Dedicated Electrical Tooling:** Pre-configured symbol tools for Branch Power (Duplex/GFCI/Quad), Switches & Dimmers, Lighting Fixtures (2x4 Troffers, Can Lights, Highbays), Distribution Gear (Panels, Transformers, J-Boxes), and Fire Alarm.
- Conduit & Cable Measurement:** Drag-to-measure tool for linear conduit runs (EMT, PVC, MC Cable, Romex NM-B) with auto-calculated footage.

B. Electrical AI Vision Recognition ([/api/takeoff/autoscan](#), [smart-scan](#))

- Electrical Symbol Recognition:** Custom Gemini Vision prompts tuned specifically for electrical legend symbols, outlet drops, switches, panel locations, and fixture counts.
- Batch Processing Optimization:** Requests queued in batches to eliminate rate limits on large multi-sheet electrical drawing sets.

C. Electrical Material Pricing & Assembly Matrix ([/estimate](#))

- Electrical Material Catalog:** National average pricing for conduit, copper wire, boxes, devices, lighting fixtures, and panel gear.
- Electrical Dependency Rules:** Automatically adds mud rings, locknuts, set-screw connectors, and assembly labor when primary devices or conduits are added.

3. Technical Backlog: Electrical Focus Roadmap

To maximize value for electrical contractors, the engineering backlog focuses exclusively on electrical trade workflow enhancements:

Electrical Module / Feature	Technical Scope & Key Objectives	Priority	Status
Panel Schedule OCR Parser	Extract circuit breaker amperages, wire sizes, and load totals directly from electrical panel schedules via AI OCR.	High	Planned
Conduit Fill & Wire Sizing Calculator	Automated NEC conduit fill calculation for EMT/PVC runs based on wire count (#12, #10, #8 THHN) and conduit size.	High	Planned
Single-Line Diagram Recognizer	Detect transformers, main switchboards, sub-panels, and feeder conduits from Electrical Riser / Single-Line diagrams.	Medium	In Progress
Lighting Fixture Schedule Sync	Parse Lighting Fixture Schedules (Type A, B, C) and automatically map symbol counts to catalog fixture SKUs.	High	Planned
Live Electrical Supplier API Sync	Real-time price syncing with electrical supply houses (Graybar, Rexel, Home Depot Pro) for localized wire & copper pricing.	Medium	Planned
Electrical Bid & Estimate Export	Generate branded Electrical Client Bid Proposals, Excel material schedules, and export to Accubid / QuickBooks.	Standard	Planned

4. Electrical Development Timeline & Schedule

Milestone & Phase	Timeline Window	Deliverables & Core Milestones	Estimated Duration
Phase 1: Electrical Takeoff & Legend Scan	Weeks 1 – 3 (Q3 2026)	- Electrical symbol legend scan & custom trade mapping - Scale calibration for linear conduit & MC cable runs	3 Weeks (Sprint 1-2)
Phase 2: Panel Schedules & Fixture OCR	Weeks 4 – 7 (Q4 2026)	- AI OCR parser for Panel Schedules & Lighting Fixture Schedules - Single-Line / Riser diagram feeder takeoff	4 Weeks (Sprint 3-4)
Phase 3: NEC Calculation Engine	Weeks 8 – 11 (Q4 2026)	- Automated NEC conduit fill & wire sizing calculator - Multi-tenant electrical project cloud database (PostgreSQL)	4 Weeks (Sprint 5-6)
Phase 4: Electrical Supplier APIs & Bidding	Weeks 12 – 16 (Q1 2027)	- Live distributor pricing sync (Graybar, Rexel, Home Depot) - Branded Electrical Proposal generator & Accubid/QuickBooks export	5 Weeks (Sprint 7-9)

5. Technical Risks & Recommended Next Steps

- Symbol Variance:** Standardize prompt templates to handle variations between IEEE/ANSI electrical blueprint symbols across architect sets.
- Volatile Commodity Prices:** Implement daily or real-time copper/wire price sync adapters so material cost estimates reflect live market rates.
- Immediate Priority:** Complete Phase 1 electrical symbol calibration and panel schedule OCR parser implementation.

Report generated automatically by Antigravity AI Pair Programmer. Project workspace: shadow-estimator. Focus: Electrical Trade.